Supplementary Material: Geometric Decomposition of Chronic Stroke Registration Asymmetry

Anand A. Joshi

1 Complete predictive-model results

Table S1 reports every evaluated and sensitivity feature family. MAE intervals were obtained by resampling participants after averaging each participant's absolute error across 20 complete outer-CV repetitions. Intervals across repeated splits are not treated as inferential confidence intervals.

Table S1: Performance of all prediction models on the common 210-participant cohort.

Model	MAE (95% CI)	RMSE	R^2	Pearson r
Intercept only	24.14 (22.28, 25.97)	27.68	-0.009	-0.119
Clinical	24.23 (22.36, 25.99)	27.83	-0.020	-0.132
Conventional lesion	16.42 (14.87, 18.09)	20.36	0.454	0.675
Lesion + deformation	16.04 (14.46, 17.72)	20.33	0.456	0.677
Deformation + registration QC	16.00 (14.33, 17.68)	20.40	0.452	0.674
Lesion + uncertainty	15.71 (14.14, 17.36)	20.11	0.468	0.688
Uncertainty + deformation	15.77 (14.23, 17.46)	19.99	0.473	0.689
Lesion + log-velocity Hodge	16.48 (14.88, 18.15)	20.50	0.447	0.670
Deformation + log-velocity Hodge	16.12 (14.57, 17.79)	20.35	0.454	0.676

Table S2 reports every prespecified paired comparison, including secondary comparisons whose intervals include zero. A positive mean advantage indicates lower repeat-averaged absolute error for the comparison model. The "Improved" column is the percentage of participants with a positive individual advantage; it is descriptive and is not the superiority criterion. Holm p values apply to the comparison family encoded in the frozen result table and do not replace the participant-bootstrap interval for the mean advantage.

Table S2: Complete paired MAE comparisons in the full cohort.

Reference	Comparison model	n	Mean advantage (95% CI)	Improved	Holm p
Conventional lesion	Clinical	210	-7.81 (-9.88, -5.71)	31.9%	< 0.001
Conventional lesion	Lesion + deformation	210	0.38 (-0.50, 1.17)	60.5%	0.046
Conventional lesion	Deformation + registration QC	210	0.43 (-0.44, 1.26)	58.6%	0.046
Conventional lesion	Lesion + uncertainty	210	0.71 (-0.10, 1.49)	61.9%	0.012
Conventional lesion	Uncertainty + deformation	210	0.65 (-0.14, 1.43)	57.6%	0.033
Conventional lesion	Intercept only	210	-7.72 (-9.75, -5.68)	31.0%	< 0.001
Conventional lesion	Lesion + log-velocity Hodge	210	-0.06 (-0.42, 0.29)	54.8%	0.688
Conventional lesion	Deformation + log-velocity Hodge	210	0.30 (-0.53, 1.08)	58.6%	0.107
Lesion + uncertainty	Uncertainty + deformation	210	-0.06 (-0.77, 0.61)	52.9%	0.723
Lesion + deformation	Deformation + log-velocity Hodge	210	-0.08 (-0.35, 0.21)	49.0%	0.573

The direct incremental mean MAE advantage of deformation after lesion uncertainty was -0.06 points (95% participant-bootstrap interval -0.77 to 0.61). The direct incremental advantage of log-velocity Hodge features after displacement features was -0.08 points (-0.35 to 0.21). Neither interval excluded zero.

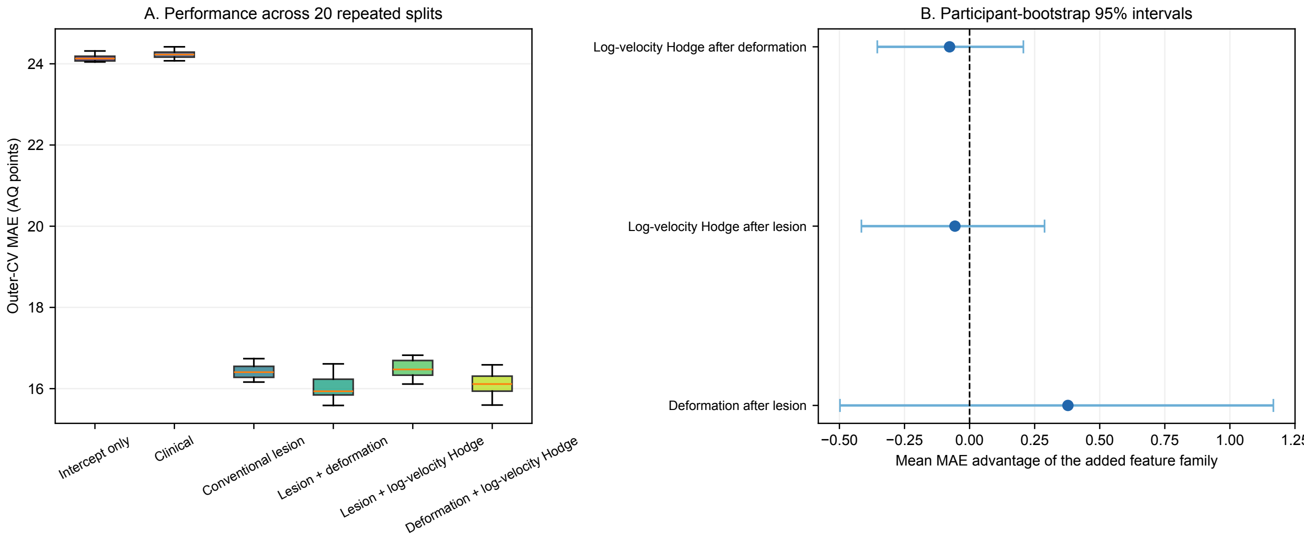

Figure S1: **Repeated nested-CV model comparison.** (A) MAE across 20 complete outer-CV repetitions. (B) Participant-level mean MAE advantages for the added feature family. Positive values favor the augmented model; bars are participant-bootstrap 95% intervals. Every incremental interval shown crosses zero.

2 Complete descriptive-association family

The 12 correlations in Table S3 formed one descriptive multiplicity family. Intervals are participant-bootstrap 95% intervals; p values are Holm adjusted across all rows. These are bivariate descriptions, not covariate-adjusted effects or causal estimates.

Table S3: Complete family of descriptive Spearman correlations.

Association	n	Spearman ρ	95% CI	Holm p
Lesion volume vs AQ	210	-0.698	(-0.769, -0.612)	< 0.001
Deformation magnitude vs lesion volume	210	0.407	(0.281, 0.528)	< 0.001
Absolute radial deformation vs lesion volume	210	0.432	(0.304, 0.543)	< 0.001
Magnitude vs registration sensitivity	210	0.399	(0.269, 0.517)	< 0.001
Absolute radial vs registration sensitivity	210	0.387	(0.256, 0.501)	< 0.001
Deformation magnitude vs AQ	210	-0.352	(-0.470, -0.222)	< 0.001
Absolute radial deformation vs AQ	210	-0.368	(-0.492, -0.239)	< 0.001
Signed radial deformation vs AQ	210	-0.235	(-0.356, -0.112)	0.002
Log-velocity RMS vs lesion volume	210	-0.021	(-0.167, 0.116)	0.759
Log-velocity RMS vs AQ	210	0.071	(-0.063, 0.206)	0.613
Curl-free energy fraction vs AQ	210	-0.240	(-0.365, -0.104)	0.002
Divergence-free energy fraction vs AQ	210	0.176	(0.038, 0.303)	0.032

3 Conventional-feature-adjusted associations

Table S4 reports partial Spearman correlations after both the exposure and AQ ranks were residualized on ranked age at stroke, log WAB assessment delay, total lesion volume, left and right language-network lesion burden, and lesion laterality. Intervals resample participants. Residual-permutation p values are

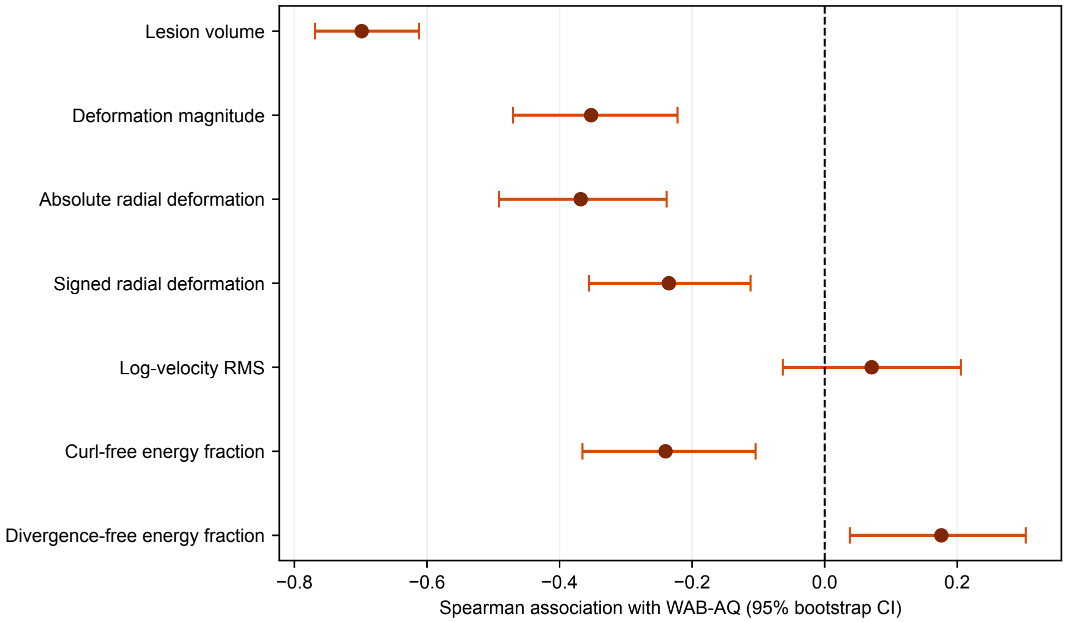

Figure S2: **Unadjusted associations with WAB-AQ.** Points are Spearman correlations and bars are participant-bootstrap 95% intervals. Multiplicity was controlled across the complete 12-test family in Table S3.

two-sided and Holm-adjusted across the six rows. No adjusted association survived that multiplicity correction.

Table S4: Exploratory partial Spearman associations with WAB-AQ.

Exposure	n	Partial ρ	95% CI	Holm permutation p
Deformation magnitude	210	-0.136	(-0.270, 0.011)	0.186
Absolute radial deformation	210	-0.146	(-0.284, -0.001)	0.161
Signed radial deformation	210	-0.047	(-0.185, 0.088)	0.973
Log-velocity RMS	210	0.012	(-0.141, 0.166)	0.973
Curl-free energy fraction	210	-0.159	(-0.290, -0.028)	0.116
Divergence-free energy fraction	210	0.138	(0.011, 0.261)	0.186

4 Log-domain numerical audit

The same numerical construction was used for all 214 cases: a 4-mm analysis grid, a raised-cosine taper over 4 voxels from the valid-domain boundary, Gaussian smoothing with $\sigma = 2.5$ voxels, and 6-voxel zero padding. The stationary logarithm used six scaling-and-squaring steps, no more than six symmetric residual updates, and a maximum relative reconstruction RMSE of 0.02. All 214 cases passed. In the analysis cohort, the median minimum input Jacobian was 0.761 (IQR 0.695–0.806), and median relative exponentiation RMSE was 0.011 (IQR 0.010–0.015).

The Fourier projection was evaluated on the entire padded grid. Nonzero frequencies were projected parallel and perpendicular to their wave vector; the zero-frequency coefficient was retained as the harmonic component. Component energies reconstructed total energy to floating-point precision in every case. Because Hodge components on bounded domains depend on boundary conditions, the reported fractions apply only to this stated tapered periodic construction.

5 Hodge parameter sensitivity

Eight one-factor perturbations varied smoothing, taper, padding, and grid resolution around the primary setting. Table S5 gives log-domain QC, rank stability relative to the primary descriptor, and the unadjusted AQ correlations. The smallest stability correlation was 0.991. Smoothing at 8 mm passed 207/214 cases, a 12-mm taper passed 212/214, and a 5-mm grid passed 213/214; the remaining variants passed all 214 cases. Component–AQ directions were unchanged, but these checks do not make the components anatomical or mechanical quantities.

Table S5: One-factor sensitivity of log-velocity/Hodge descriptors. Stability columns are Spearman correlations with the primary descriptor; Curl–AQ and Div–AQ are unadjusted Spearman correlations.

Variant	QC	n	RMS stability	Curl stability	Curl–AQ	Div–AQ
Smoothing $\sigma = 8$ mm	207/214	203	0.997	0.994	-0.263	0.206
Smoothing $\sigma = 12$ mm	214/214	210	0.998	0.995	-0.220	0.149
Taper 12 mm	212/214	208	0.991	0.994	-0.255	0.181
Taper 20 mm	214/214	210	0.993	0.996	-0.223	0.165
Padding 16 mm	214/214	210	1.000	1.000	-0.248	0.168
Padding 32 mm	214/214	210	1.000	1.000	-0.237	0.184
Grid 3 mm	214/214	210	0.998	0.996	-0.231	0.168
Grid 5 mm	213/214	209	0.997	0.996	-0.238	0.175

6 Outcome-scope audit

WAB aphasia type was unavailable for 29 of 210 participants. Available counts were Broca 77, anomic 57, conduction 25, global 13, Wernicke 5, and transcortical motor 4. Because type is derived from the WAB examination used to define AQ, and because the rare classes do not support a credible held-out classifier, subtype was not introduced as a second endpoint. The frozen joined table contained no independent language-task outcome suitable for an additional analysis.

7 Left-dominant-lesion sensitivity analysis

Excluding the single right-dominant case left $n = 209$. The mean MAE advantage of displacement features was 0.33 points (95% participant-bootstrap interval -0.51 to 1.09), again failing the superiority criterion. Table S6 reports the full left-dominant-only comparison family; none of the incremental deformation or Hodge intervals excluded zero.

Table S6: Paired MAE comparisons after excluding the single right-dominant lesion case.

Reference	Comparison model	n	Mean advantage (95% CI)	Improved	Holm p
Conventional lesion	Lesion + deformation	209	0.33 (-0.51, 1.09)	58.4%	0.061
Conventional lesion	Lesion + log-velocity Hodge	209	-0.02 (-0.35, 0.30)	53.1%	0.630
Conventional lesion	Deformation + log-velocity Hodge	209	0.30 (-0.57, 1.11)	58.9%	0.103
Lesion + deformation	Deformation + log-velocity Hodge	209	-0.03 (-0.29, 0.23)	50.7%	0.950

8 Interpretation boundary

The stationary vector field is a group-log registration parameter with displacement units over an arbitrary unit interval. It is not the original time-dependent velocity optimized by LDDMM and is not physical tissue velocity. The curl-free component, divergence, and log-Jacobian are geometric quantities; without a constitutive model, tissue parameters, loads, and mechanical boundary conditions, none is an estimator of pressure.